

Derivative of a function (EGE Mathematics, grade 11)

The derivative $f'(x)$ measures the instantaneous rate of change of $f(x)$.

For a power function $f(x) = x^n$, the derivative is $f'(x) = n \cdot x^{n-1}$.

The derivative of $\sin(x)$ is $\cos(x)$; the derivative of $\cos(x)$ is $-\sin(x)$.

At an extremum point the derivative equals zero: $f'(x) = 0$.

Increasing intervals correspond to $f'(x) > 0$, decreasing to $f'(x) < 0$.

Example EGE task: find the point of maximum of $y = x^3 - 3x$.

Solution: $y' = 3x^2 - 3 = 0 \Rightarrow x = -1$ gives the maximum.